

Boundary, cosmological, and observational extensions to the Systrophē framework

Five further modules (Phases 16–20)

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Abstract

We add five further modules to the Systrophē framework [?, ?, ?, ?] covering boundary, cosmological, and observational aspects of the Tipler / Lewis-Papapetrou structure: (1) a speculative AdS/CFT-style dictionary mapping the supercritical exterior to a non-unitary CFT with complex conformal weights $\Delta = 1 \pm i\alpha/2$; (2) quantum-information channel capacities for radial transport in the LP exterior, including Deutsch and Lloyd CTC channel models; (3) cosmological embedding of the cylinder in an FRW background, yielding a Hubble-induced "evaporation" of CTC bands; (4) spinor monodromy along closed phi-loops with explicit Spin(1,3) holonomy matrices and a Systrophē-pair extinction proxy; (5) synchrotron- analog radiation from charged orbits in CTC bands. Eighty-two new tests across the five modules, all passing. Code at github.com/Zynerji/systrophe (v0.18.0+).

1 Holographic dictionary (Phase 16)

The LP supercritical exterior is not asymptotically AdS, but at large r the metric components oscillate log-periodically about a power-law envelope. Treating this as a "log-periodic CFT" boundary, we extract effective conformal weights

$$\Delta = 1 \pm i\alpha/2,$$

which are *complex*: a hallmark of non-unitary CFTs with discrete-scale-invariant RG flow [?, ?].

The module `holographic.py` provides:

- `boundary_expansion_coefficients`: extract (β, α, δ) at large r ;
- `dual_operator_dimensions`: complex Δ from α ;
- `brown_york_stress_tensor`: quasi-local stress at finite r ;
- `holographic_entropy_strip`: Ryu-Takayanagi-style entanglement;
- `boundary_two_point_correlator`: $\langle O(s)O(0) \rangle \sim s^{-2}e^{-i\alpha \ln s}$;
- `efimov_correspondence`: map to 3-body Efimov scaling λ_E .

A two-point correlator phase fit recovers α to within 1% of the theoretical value (test `test_reconstruct_alpha_matches_theoretical`).

2 Quantum-information channel (Phase 17)

The radial worldline in the LP exterior is treated as a noisy quantum channel. Each chronology-horizon crossing applies an amplitude-damping with parameter p_{CH} ; the cumulative damping after N crossings is $1 - (1 - p_{\text{CH}})^N$. We adopt the heuristic classical capacity

$$C(p) = \max \left[0, 1 - h_2 \left(\frac{1 + \sqrt{1-p}}{2} \right) \right],$$

with h_2 the binary entropy.

The module `qi_channel.py` provides:

- `redshift_factor` and `amplitude_damping_p`;
- `channel_capacity_holevo`: classical capacity;
- `chronology_horizon_decoherence`: cumulative p after N CHs;
- `closed_timelike_loop_capacity`: Deutsch classical / quantum capacities (both 0) and Lloyd P-CTC capacity (1 bit);
- `entanglement_swap_protocol`: Bell pair fidelity vs. radius;
- `channel_assisted_clock_sync`: time-dilation residual.

Headline: the classical Deutsch capacity of a CTC channel vanishes identically (the self-consistent fixed point erases input), but the Lloyd P-CTC capacity gives 1 bit per qubit. This distinction is a testable consequence of the chosen CTC model.

3 Cosmological embedding (Phase 18)

A cylinder embedded in an FRW background is exposed to Hubble drag. We model the drag as an effective rotation reduction $\omega_{\text{eff}}(t) = \omega(1 - H(t)/\omega)$, which propagates to a corrected log-frequency $\alpha_{\text{eff}} = \sqrt{4a_{\text{eff}}^2 - 1}$. For Λ -dominated FRW with sufficiently large H_0 , a_{eff} drops below $1/2$ and the cylinder *evaporates*: all CTCs are extinguished.

The module `cosmological_embedding.py` provides:

- `frw_scale_factor` and `hubble_parameter` for matter / radiation / Λ EoS;
- `matching_residual`: LP vs. FRW mismatch at r_{match} ;
- `cosmological_corrections_to_alpha`: shifted log-frequency;
- `horizon_chirp_rate`: dr_1/dt for the first CH;
- `cylinder_evaporates_under_expansion`: evaporation time;
- `inflation_creates_systrophe_cylinders`: spontaneous creation rate during inflation, $H^4 e^{-S_E}$.

This connects the Systrophē framework to cosmological observability: a primordial Systrophē cylinder could survive matter / radiation domination but evaporates in late-time Λ domination.

4 Spinor monodromy along CTCs (Phase 19)

A Dirac spinor parallel-transported around a closed phi-loop at constant t and r in the LP exterior picks up a $\text{Spin}(1,3)$ holonomy

$$U(r) = \exp\left[-\frac{\pi}{2} \omega_\phi^{ab}(r) \gamma_a \gamma_b\right].$$

The module `spinor_monodromy.py` provides:

- `spin_connection_phi`: $\omega_\phi^{ab}(r)$ via finite-difference;
- `spinor_holonomy`: 4×4 holonomy matrix;
- `monodromy_eigenvalues`: spectrum of U ;
- `fixed_point_spinors`: eigenvectors at $+1$;
- `multi_loop_monodromy`: U^N ;
- `pair_modified_monodromy`: δ -interpolation between U and identity (Systrophē-pair extinction);
- `chronology_horizon_caustic`: blow-up loci of $|\omega_\phi^{ab}|$.

At $\delta = \pi$ (full pair extinction), the pair monodromy collapses to the identity (test `test_pair_modified_monodromy`). This echoes the classical L_{pair} extinction proxy of Phase 4 and shows that the chronology-protection mechanism extends consistently to spinor representations.

5 Synchrotron radiation analog (Phase 20)

A charged particle on a circular orbit in a CTC band radiates electromagnetically. The orbit frequency $\Omega(r)$ follows from the geodesic condition $F' + 2\Omega K' + \Omega^2 L' = 0$; the effective Lorentz factor is $\gamma^{-2} = F - 2\Omega K - \Omega^2 L$; the synchrotron critical frequency is $\omega_c = \frac{3}{2}\gamma^3\Omega$.

The module `synchrotron_analog.py` provides:

- `orbital_frequency`, `effective_gamma_factor`;
- `synchrotron_critical_frequency`;
- `emitted_power`: Larmor-like $P = (2/3)e^2 a^2 \gamma^4$;
- `characteristic_emission_band`;
- `multi_band_synchrotron_spectrum`: emission per CTC band;
- `back_reaction_drag`;
- `observable_signature_distant_observer`: redshift to far observer.

Headline: each CTC band emits a synchrotron-like spectrum with peak at ω_c set by r . A distant observer sees a series of discrete log-periodically-spaced emission bands, redshifted by $\sqrt{F_{\text{emit}}/F_{\text{obs}}}$. This is a direct astrophysical signature – if a Systrophē cylinder existed, its CTC bands would glow.

6 Implications

The five modules collectively extend the Systrophē framework into:

1. **Holographic / RG-cycle territory:** the supercritical exterior is a concrete realization of a log-periodic CFT with complex conformal weights.
2. **Quantum-information physics:** chronology horizons function as amplitude-damping channels, with Deutsch and Lloyd CTC models giving distinct testable capacities.
3. **Cosmological context:** cylinders evaporate under Λ expansion; inflation creates them at rate $H^4 e^{-S_E}$.
4. **Spinorial physics:** explicit 4×4 holonomy matrices for parallel transport; pair extinction extends to spinor representations.
5. **Observational astrophysics:** synchrotron-like emission from CTC bands gives a red-shifted log-periodic emission spectrum that a distant observer could detect.

7 Reproduction

Phase	Source file
16. Holographic dictionary	<code>holographic.py</code>
17. QI channel	<code>qi_channel.py</code>
18. Cosmological embedding	<code>cosmological_embedding.py</code>
19. Spinor monodromy	<code>spinor_monodromy.py</code>
20. Synchrotron analog	<code>synchrotron_analog.py</code>

All 82 new tests pass; 742-test full suite passes end-to-end.

References

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